A.6.2 Plotting Data with Error Bars

The built-in *Mathematica* function `ListPlot` allows you to plot data points, but gives you no way to include errors bars for each point.

This package defines the function `DataPlot`, which is an analog of `ListPlot` for data with error bars.

```
DataPlot::usage =
        "DataPlot[list] plots a list of data with error bars. The
        data can be given either in the form {{y1, dy1}, {y2, dy2}, ...}
        or {{x1, y1, dy1}, ...}."

DataPlot[l2:{{_, _}..}] :=
        Block[ {i}, DataPlot[ Table[Prepend[l2[[i]], i], {i, Length[l2]}] ] ]

DataPlot[l3:{{_, _, _}..}] :=
        Show[ Graphics[ { PointSize[0.015], Thickness[0.002],
                Block[ {i, x, y, dy} ,
                Table[
                        {x, y, dy} = l3[[i]] ;
                        { Line[ {{x, y-dy}, {x, y+dy}} ],
                        Point[ {x, y} ] } ,
                        {i, Length[l3]}
                ] ] } ], Axes -> Automatic ]
```

Plotting data with error bars.

Notice the use of pattern objects such as `{_,_}..` to recognize sequences of lists of two elements.

The basic way that `DataPlot` works is to construct a graphics object which is then displayed using `Show`.

Read in the package.	*In[1]:=* **`<<DataPlot.m`**
This makes a list of 10 pairs of random numbers, with the second number in the pair on average a factor of ten smaller than the first.	*In[2]:=* **`data = Table[{Random[Real, 10], Random[Real, 1]}, {10}] ;`**

Web sample page from The *Mathematica* Book, First Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-19334-5; softcover ISBN 0-201-19330-2). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; http://www.wolfram.com/; 1-800-441-6284.

Here is the list `data`, with the first number in each pair interpreted as y position, and the second number as y error.

In[3]:= **`DataPlot[%]`**